



BRI-Adjusted Floor Level Model

An Additive Flood-Threshold Credit Earned from Building Resilience

MKM Research Labs

Version 1.0 — 21-July-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs (“MKM”).

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an “as is” basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

Contents

1	Executive Summary	3
2	Model Purpose and Scope	3
2.1	Purpose	3
2.2	Scope	4
2.3	Out of Scope	4
3	Mathematical Framework	4
3.1	The Uplift Ramp	4
3.2	Adjusted Floor Level	4
3.3	Anchor Placement	4
3.4	Letter-Grade Fallback	5
3.5	Relationship to the Damage-Curve Stilt	5
4	Input Parameters and Calibration	5
4.1	Calibration Status	5
4.2	Data Inputs	6
5	Implementation Details	6
5.1	Consumer Contract	6
6	Sensitivity Analysis	6
7	Validation and Backtesting	8
7.1	Verification Approach	8
7.2	Backtesting	9
7.3	Portfolio-Level Check	9
8	Model Limitations and Known Weaknesses	9
8.1	Remediation	10
9	Governance Validation Questions	10
9.1	Q1 — Model purpose and use	10
9.2	Q2 — Conceptual soundness	10
9.3	Q3 — Data quality	10
9.4	Q4 — Assumptions	11
9.5	Q5 — Implementation verification	11
9.6	Q6 — Sensitivity and stability	11
9.7	Q7 — Limitations and weaknesses	11
9.8	Q8 — Ongoing monitoring	11
9.9	Q9 — Governance and approval	11
10	Change History	11

1 Executive Summary

The BRI-Adjusted Floor Level Model (MKM-BRF-001) answers a narrow but consequential question: at what water level does a property actually flood, once its evidenced flood resilience is taken into account? It converts the Building Resilience Index flood sub-score (MKM-BRI-001) into an additive floor-level credit in metres, and adds that credit to the surveyed floor level to give the effective flood threshold that every downstream consumer compares against the attenuated water-surface elevation.

The model is small by design — a monotone ramp and an addition — but it sits on the critical path of the Property Risk Spread. Because the PRS flood leg is an *event count*, and because the event count is decided by exactly this threshold comparison, the credit this model awards translates directly into basis points. A property that clears the threshold in fewer of the 10,000 simulated storms earns a narrower spread.

Two properties of the credit are deliberate and material:

- **It is additive, never subtractive.** A poor BRI score awards zero credit; it can never lower the effective threshold below the surveyed floor. Below-reference vulnerability is expressed elsewhere — through the signed *stilt* on the damage curve (MKM-DD-001) — so that vulnerability and resilience enter through separate, auditable channels rather than being netted inside one number.
- **It is banded, not continuous from zero.** No credit is awarded at or below a flood sub-score of 0.38 (the NR/B boundary); the full 3 m is awarded at or above 0.87 (the AA anchor); the credit ramps linearly in between. Resilience must be evidenced to a threshold standard before it earns anything.

Status: Development (Tier 2, RAG Amber). The uplift ramp, the adjusted floor level and the letter-grade fallback are delivered with unit tests. The three anchors (`MAX_M`, `SCORE_LO`, `SCORE_HI`) are engineering-judgement seeds.

Key risk: A 3 m maximum uplift is very large relative to typical flood depths in the modelled catchments. As the sensitivity analysis shows, an AA-graded property with a 0.30 m surveyed floor does not flood until the water is more than 3.3 m above ground — which suppresses its flood event count to near zero. The credit is therefore close to a binary exemption for top-graded properties, and the anchor has no empirical basis. See BRF-L1.

2 Model Purpose and Scope

2.1 Purpose

To reward evidenced flood resilience in the flood threshold itself rather than only in the damage curve. Physical flood-resilience measures — raised thresholds, flood doors, non-return valves, tanked basements, demountable barriers — act primarily by keeping water out, not by reducing damage once water is in. Modelling them purely as a damage-curve adjustment would misplace the mechanism; this model puts the credit where the physics is.

2.2 Scope

In scope: the mapping from BRI flood sub-score to a floor-level uplift in metres; the addition of that uplift to the surveyed floor level; and the representative-score fallback for assets that carry a BRI letter grade instead of a numeric sub-score.

2.3 Out of Scope

- Producing the BRI flood sub-score — owned by MKM-BRI-001.
- The damage curve and its signed stilt — owned by MKM-DD-001.
- The water-surface elevation and its attenuation — owned by the flood propagation and spatial models.
- Surveying the floor level. `FloorLevelMeters` is a CDM input.
- Any assessment of whether the resilience measures are correctly installed, maintained or operable at the time of an event.

3 Mathematical Framework

3.1 The Uplift Ramp

Let $B_f \in [0, 1]$ be the BRI flood sub-score, with anchors $s_{lo} = 0.38$, $s_{hi} = 0.87$ and maximum uplift $U_{max} = 3.00$ m. The credit is

$$U(B_f) = \begin{cases} 0, & B_f \leq s_{lo}, \\ \frac{B_f - s_{lo}}{s_{hi} - s_{lo}} U_{max}, & s_{lo} < B_f < s_{hi}, \\ U_{max}, & B_f \geq s_{hi}. \end{cases}$$

The score is clamped to $[0, 1]$ before evaluation. U is continuous, non-decreasing and bounded — three properties the unit tests assert directly.

3.2 Adjusted Floor Level

$$F_{adj} = F_{surveyed} + U(B_f) \geq F_{surveyed}.$$

Consumers compare `GroundLevelMeters` + F_{adj} against the attenuated water-surface elevation. The property floods when the water clears the adjusted level.

3.3 Anchor Placement

The two score anchors are not arbitrary: they are the boundaries of the BRI letter-grade bands ($AA \geq 0.87$, $A \geq 0.62$, $B \geq 0.38$, $NR < 0.38$). Placing s_{lo} at the NR/B boundary means an unrated property earns nothing; placing s_{hi} at the AA anchor means the full credit requires top-band evidence. A property at the B/A boundary (0.62) earns $(0.62 - 0.38)/(0.87 - 0.38) \times 3.00 = 1.47$ m, just under half.

3.4 Letter-Grade Fallback

Commercial assets record flood resilience as a Water/Flash grade envelope rather than the residential numeric sub-score. For these, a representative score is taken as the mid-point of the grade’s band:

$$AA \rightarrow 0.935, \quad A \rightarrow 0.745, \quad B \rightarrow 0.500, \quad NR \rightarrow 0.190,$$

and the continuous ramp is evaluated there. A trailing + modifier is stripped before lookup. An unknown, absent or “N/A” grade returns no representative score — the caller decides how to treat a non-applicable grade, rather than the model silently substituting one.

Note the asymmetry this creates: a B-graded commercial asset evaluates at 0.500 and earns 0.73 m of credit, whereas an NR-graded one evaluates at 0.190 and earns nothing, because 0.190 sits below s_{lo} .

3.5 Relationship to the Damage-Curve Stilt

The two BRI channels are complementary and must not be confused:

Table 1: The two BRI flood channels

	Floor uplift (this model)	Stilt (MKM-DD-001)
Sign	non-negative only	signed
Cap	+3.00 m	± 1.00 m
Form	piecewise-linear ramp	log-differential
Acts on	the flood <i>trigger</i>	the <i>damage</i> given flooding
Drives	PRS event count	loss severity

A property with a poor score is penalised through a negative stilt, not through a negative uplift. A property with a strong score is rewarded through both — it floods less often *and* takes less damage when it does. This double credit is intentional but should be understood when attributing spread movements.

4 Input Parameters and Calibration

Table 2: Model parameters (`config/bri.py`)

Symbol	Constant	Value	Meaning
$U_{\max}$	BRI_FLOOR_UPLIFT_MAX_M	3.00	maximum credit (m)
s_{lo}	BRI_FLOOR_UPLIFT_SCORE_LO	0.38	no-credit boundary
s_{hi}	BRI_FLOOR_UPLIFT_SCORE_HI	0.87	full-credit anchor
—	BRI_FLOOR_RATING_SCORES	see §3.4	grade mid-points

4.1 Calibration Status

The anchors are engineering-judgement seeds. s_{lo} and s_{hi} are defensible independently of loss data because they are defined by the BRI grade bands, which are themselves the reviewed output of

MKM-BRI-001. $U_{\max}$ is not: 3 m is a judgement about how much water a comprehensively-protected building can exclude, and it should be recalibrated against post-event resilience performance as soon as loss data allows. This is remediation item BRF-R1.

4.2 Data Inputs

Table 3: Per-property inputs

Input	Source	Missing-value behaviour
FloorLevelMeters	property CDM	caller-supplied; no default here
BRIFloodScore	resilience CDM	—
BRI flood letter grade	commercial resilience CDM	None if unmappable

5 Implementation Details

The model is implemented alongside the flood damage curve in `src/models/floodrisk/depth_damage.py`, which is the natural home because the two share the BRI inputs and are always configured together.

Table 4: Public functions

Function	Returns
<code>bri_floor_uplift(score)</code>	credit in metres, $[0, U_{\max}]$
<code>bri_adjusted_floor_level(floor, score)</code>	adjusted level in metres
<code>flood_rating_to_score(rating)</code>	representative score or <code>None</code>

All three are pure functions of their arguments — no configuration is read at call time beyond the module-level constants, no I/O is performed, and no state is retained. This makes the model trivially testable and means a change to the anchors takes effect uniformly on the next process start.

5.1 Consumer Contract

Consumers must apply the adjusted floor level to the *trigger* comparison only. Passing F_{adj} into `bri_depth_damage` as the `floor_level` argument would apply the uplift a second time, once through the trigger and once through the effective-depth subtraction, and would also compound with the stilt. The damage curve takes the *surveyed* floor level.

6 Sensitivity Analysis

Generated by calling the model directly (`python -m docs.models.sensitivities.generate_all --model brifloor`).

The material observations:

- **The credit dominates the surveyed floor level.** Typical surveyed floors span 0 to 1.5 m; the credit spans 0 to 3 m. For an AA property the credit is the dominant term in the effective threshold by a factor of two or more.

- **The trigger table is the model’s real output.** For a 0.30 m surveyed floor, NR floods at 0.5 m of water while AA needs more than 3.3 m. Across a storm set where most events produce sub-metre depths, this is close to an exemption for AA properties — the mechanism behind BRF-L1.
- **Grade granularity is coarse at the bottom.** NR (0.190) and any score below 0.38 all earn exactly zero, so the model cannot distinguish degrees of poor resilience. That distinction is left to the stilt.
- **Residual damage falls steeply with grade.** At 1.5 m of water the baseline curve gives materially different damage across grades purely because the depth above the adjusted floor differs.

Table 5: Floor-level uplift (m) by BRI flood sub-score. No credit at or below 0.38, the full 3.0 m at or above 0.87, linear in between.

BRI flood score	Uplift (m)	Band
0.0000	0.0000	no credit
0.1900	0.0000	no credit
0.3000	0.0000	no credit
0.3800	0.0000	no credit
0.5000	0.7350	linear ramp
0.6200	1.47	linear ramp
0.7450	2.23	linear ramp
0.8700	3.00	full credit
0.9350	3.00	full credit
1.00	3.00	full credit

Table 6: Adjusted floor level (m) by surveyed floor level and BRI grade mid-point score. The adjustment is additive, so the adjusted level never falls below the surveyed level.

Surveyed floor (m)	NR (0.190)	B (0.500)	A (0.745)	AA (0.935)
0.0000	0.0000	0.7350	2.23	3.00
0.1500	0.1500	0.8850	2.38	3.15
0.3000	0.3000	1.03	2.54	3.30
0.6000	0.6000	1.33	2.83	3.60
1.00	1.00	1.74	3.23	4.00
1.50	1.50	2.23	3.73	4.50

Table 7: Letter-grade fallback: representative flood score and resulting uplift. A trailing + modifier is stripped before lookup; unknown or non-applicable grades return no representative score.

BRI rating	Representative score	Uplift (m)
AA	0.9350	3.00
AA+	0.9350	3.00
A	0.7450	2.23
A+	0.7450	2.23
B	0.5000	0.7350
NR	0.1900	0.0000
N/A	not mappable	n/a
ZZ	not mappable	n/a

Table 8: Flood trigger by BRI grade for a property with a 0.3 m surveyed floor level. The credit raises the depth at which the property first floods, which is how the model suppresses PRS flood event counts.

Water above ground (m)	NR grade	B grade	A grade	AA grade
0.2500	no	no	no	no
0.5000	yes	no	no	no
1.00	yes	no	no	no
1.50	yes	yes	no	no
2.00	yes	yes	no	no
3.00	yes	yes	yes	no
4.00	yes	yes	yes	yes

Table 9: Damage ratio through the baseline depth-damage curve, evaluated at the depth above the adjusted floor level. The gap between NR and AA at a given water level is the resilience credit expressed as avoided loss.

Water above ground (m)	NR grade	B grade	A grade	AA grade
0.2500	0.0000	0.0000	0.0000	0.0000
0.5000	0.1167	0.0000	0.0000	0.0000
1.00	0.3100	0.0000	0.0000	0.0000
1.50	0.4400	0.2346	0.0000	0.0000
2.00	0.5400	0.3896	0.0000	0.0000
3.00	0.7050	0.5931	0.2346	0.0000
4.00	0.8200	0.7448	0.4931	0.3100

7 Validation and Backtesting

7.1 Verification Approach

Verified by unit test against calibration-independent properties:

- **Bounds.** $U \in [0, U_{\max}]$ for every input including out-of-range and negative scores, which are clamped.

- **Monotonicity.** U is non-decreasing in the score.
- **Anchor exactness.** $U(s_{lo}) = 0$ and $U(s_{hi}) = U_{\max}$ exactly, with continuity across both boundaries.
- **Non-subtractivity.** $F_{\text{adj}} \geq F_{\text{surveyed}}$ for every score, including zero.
- **Grade mapping.** Each letter maps to its band mid-point; the + modifier is stripped; unknown, empty, None and “N/A” return None rather than a default score.

7.2 Backtesting

No backtest has been performed. A meaningful test requires paired observations of resilience grade, water level and whether the property actually took water in — available only from post-event survey. This is remediation item BRF-R1 and is a precondition of MRC sign-off.

7.3 Portfolio-Level Check

The most informative available check is the distribution of flood event counts by BRI grade across the 10,000-storm run. A monotone decline in mean event count from NR to AA is the expected signature; a collapse to zero at AA would confirm BRF-L1 is binding in practice and would argue for reducing $U_{\max}$.

8 Model Limitations and Known Weaknesses

BRF-L1 — The maximum uplift may act as an exemption. $U_{\max} = 3$ m exceeds the flood depth of most simulated events. Top-graded properties may therefore record near-zero flood event counts and near-zero flood spread, which is a much stronger statement than the underlying evidence supports. *Mitigation:* monitor the event-count distribution by grade; recalibrate $U_{\max}$ under BRF-R1.

BRF-L2 — Resilience measures are assumed always operable. Demountable barriers must be deployed, non-return valves must not be blocked, flood doors must be closed. The model credits installed capability, not realised performance, and applies no deployment-failure probability.

BRF-L3 — No depth-duration dependence. A property’s ability to exclude water degrades with the duration of inundation and with hydrostatic pressure. The credit is a fixed height regardless of how long the water stands against it.

BRF-L4 — Coarse at the low end. All scores at or below 0.38 receive identical treatment, so the model cannot rank poor properties against each other.

BRF-L5 — Grade mid-points for commercial assets. Using band mid-points for letter-graded assets understates within-grade dispersion, and the choice of mid-point rather than, say, band floor is a conservative-vs-central judgement that has not been validated.

BRF-L6 — Double credit with the stilt. A high-scoring property is rewarded twice — fewer trigger events and lower damage per event. Intentional, but it means grade sensitivity of the total spread is stronger than either channel alone would suggest.

BRF-L7 — No interaction with groundwater or sewer surcharge. The credit is applied against surface water level only; ingress paths that bypass the floor threshold are not represented.

8.1 Remediation

BRF-R1 Recalibrate $U_{\max}$ against post-event resilience performance; precondition of sign-off (addresses BRF-L1).

BRF-R2 Introduce a deployment-reliability factor for measures that require human action (addresses BRF-L2).

BRF-R3 Add the by-grade flood event-count distribution to the standard monitoring pack.

BRF-R4 Document and test the consumer contract that the damage curve receives the surveyed — not the adjusted — floor level, with an assertion at the integration boundary.

9 Governance Validation Questions

The following nine questions are mandated by Chapter 5 of the *Model Risk Governance for Vendors* handbook and must be explicitly addressed for every model in the inventory. Response status is tracked in the Model Governance Dashboard (MRC portal).

9.1 Q1 — Model purpose and use

“Converts the BRI flood sub-score into an additive floor-level credit and returns the effective flood threshold used by the flood trigger. Directly drives PRS flood event counts.”

9.2 Q2 — Conceptual soundness

“Crediting flood resilience at the trigger rather than in the damage curve matches the physical mechanism of the measures concerned. The ramp anchors are taken from the reviewed BRI grade bands rather than chosen freely. The deliberate non-negativity keeps resilience and vulnerability in separate, auditable channels.”

9.3 Q3 — Data quality

*“Inputs are the CDM surveyed floor level and the BRI flood sub-score or letter grade. The model performs no imputation: an unmappable grade returns **None** and the caller must decide. Coverage of `FloorLevelMeters` and `BRI FloodScore` should be monitored, since a missing floor level silently becomes a zero baseline at the call site.”*

9.4 Q4 — Assumptions

“Resilience measures are always operable (BRF-L2); excluded-water capability is a fixed height independent of duration (BRF-L3); grade mid-points represent letter-graded assets (BRF-L5).”

9.5 Q5 — Implementation verification

“Unit tests cover the ramp bounds, monotonicity, both anchors and their continuity, clamping of out-of-range scores, non-subtractivity of the adjusted level, and every branch of the grade-mapping fallback.”

9.6 Q6 — Sensitivity and stability

“See Section 6. The response is monotone and, at the top of the range, very strong — strong enough that BRF-L1 is raised as the model’s principal risk. The model is deterministic and closed-form; stability is not a concern.”

9.7 Q7 — Limitations and weaknesses

“Documented in Section 8 (BRF-L1–BRF-L7). BRF-L1 is the material one: an uncalibrated 3 m maximum that can suppress the flood leg of top-graded properties almost entirely.”

9.8 Q8 — Ongoing monitoring

“Not yet in production. Monitoring under BRF-R3 covers the by-grade flood event-count distribution and CDM field coverage.”

9.9 Q9 — Governance and approval

“Owner: David K Kelly. Status: Development, Tier 2, RAG Amber. Not approved for production; sign-off is contingent on closing BRF-R1–BRF-R4.”

10 Change History

Version	Date	Description
1.0	21-July-2026	Initial model documentation: uplift ramp, adjusted floor level, letter-grade fallback, r